

HPC Network Topologies

This guide explores the structure, behavior, and performance implications of different network topologies used in High-Performance Computing (HPC) environments. The activity combines conceptual understanding with analytical and practical exercises.

Part I – Conceptual Understanding

Answer the following questions concisely and accurately.

1. Define the concept of a *network topology* in the context of HPC systems.
2. Explain why network topology plays a critical role in large-scale parallel applications.
3. Compare the following topologies in terms of scalability, latency, and cost:
 - Fat-tree
 - Mesh
 - Dragonfly
4. Describe the relationship between topology design and message-passing efficiency (MPI).

Part II – Analytical and Practical Exercises

Perform the following analytical tasks. Use diagrams or tables where appropriate.

1. Given a 64 nodes mesh topology, calculate the network diameter and the number of network cables.
2. Draw a schematic of a 2-level Fat-tree network interconnecting 16 compute nodes. Label core, aggregation, and leaf switches.
3. Consider an HPC cluster of 512 nodes connected via a Dragonfly topology with 8 groups and 64 nodes per group:
 - a) Compute the total number of local and global links.
 - b) Discuss how congestion might develop under all-to-all communication.
4. Evaluate the performance impact of link failures in a Fat-tree vs. a Torus topology. Propose strategies to mitigate them.
5. Discuss the scalability challenges when extending a Dragonfly network to exascale systems.